

Avikshith Reddy

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Summary

Data Scientist specializing in customer acquisition, retention modeling, and experimentation workflows, building scalable **machine learning** systems using **Python**, **SQL**, **AWS SageMaker**, and **XGBoost** across regulated financial and customer analytics environments. Experienced across the full ML lifecycle including feature engineering, predictive modeling, uplift analysis, experimentation, MLOps, and LLM-powered analytics systems delivering measurable business and customer engagement impact.

Technical Skills

Programming & Analytics: Python, SQL, Pandas, NumPy, PySpark, scikit-learn, TensorFlow, PyTorch, Statsmodels, Tableau, Power BI

Machine Learning & AI: Classification, Regression, Clustering, Customer Analytics, Segmentation, Propensity Modeling, Churn Prediction, Recommendation Systems, Forecasting, Feature Engineering, Model Evaluation, Model Validation, Hyperparameter Tuning

Generative AI & NLP: RAG Pipelines, NLP, LangChain, LlamaIndex, Embeddings, Semantic Retrieval, HuggingFace Transformers, Vector Databases

Experimentation & Analytics: EDA, Statistical Modeling, Hypothesis Testing, Experimental Design, A/B Testing, Causal Inference, Uplift Modeling

Data Engineering: Databricks, ETL Pipelines, Distributed Data Processing, Data Modeling, Data Wrangling, Data Validation, Data Ingestion

MLOps & Deployment: MLflow, Docker, CI/CD, FastAPI, Model Deployment, Model Monitoring, Drift Detection, Git, APIs

Cloud Technologies: AWS (S3, SageMaker, Lambda, Redshift), Azure (Azure ML, Data Factory, Blob Storage)

Experience

Freddie Mac

July 2025 – Present

Data Scientist

Dallas, TX

- Developed customer-risk and propensity-focused **machine learning** models using **Python**, **SQL**, and **scikit-learn**, improving portfolio risk-classification accuracy by **14%** across regulated mortgage datasets supporting customer engagement and retention analytics.
- Optimized predictive modeling workflows using **XGBoost**, feature engineering, & hyperparameter-tuning methodologies, selecting algorithms aligned to business objectives while improving targeting precision & response analysis across financial analytics environments.
- Engineered scalable analytical pipelines using **PySpark**, **Databricks**, and distributed transformation workflows, integrating multi-source financial datasets into governed analytical layers supporting experimentation, segmentation, and next-best-action modeling initiatives.
- Operationalized monitored **MLOps** workflows using **AWS SageMaker**, **MLflow**, **Docker**, and CI/CD pipelines, enabling reproducible experimentation, controlled deployment, model evaluation, and scalable inference systems alongside data engineering teams.

Yogin

May 2021 – Aug 2023

Data Science Associate

India

- Built customer-acquisition and response-prediction **machine learning** workflows using **Python**, **XGBoost**, and behavioral feature engineering, improving lead-scoring model performance by **18%** across customer-targeting and engagement datasets.
 - Executed controlled experimentation and **uplift modeling** frameworks using **SQL**, statistical analysis, and hypothesis-testing methodologies, enabling stakeholders to optimize campaign targeting strategies through measurable customer-conversion and retention-impact evaluation workflows.
 - Constructed ETL pipelines using **PySpark**, relational databases, & transformation workflows, integrating multi-source customer datasets into scalable analytical environments supporting model training, recommendation workflows, & operational reporting initiatives.
 - Delivered stakeholder-facing analytical reporting solutions using **Tableau**, exploratory analysis, and KPI modeling, translating predictive-model outputs and customer-segmentation insights into actionable recommendations for business and marketing teams.
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AI & Data Projects

Fraud Detection MLOps System (IEEE-CIS) | Python, scikit-learn, MLflow, FastAPI, Docker, AWS

- Developed a production-grade **fraud detection pipeline** engineering behavioral features & ensemble models with leakage-aware validation, achieving **ROC-AUC 0.8949**. Deployed FastAPI inference services with MLflow tracking & automated retraining workflows.

Customer Churn Prediction and Segmentation Platform | Python, pandas, scikit-learn, XGBoost

- Built an end-to-end **customer retention analytics pipeline** that engineers behavioral & RFM features, trains churn & propensity models, achieving **ROC-AUC 0.7555**, and generates actionable customer segments supporting targeted retention & engagement strategies.

Executive Review Intelligence System (Customer Review Analytics Platform) | LlamaIndex, ChromaDB, HuggingFace, Streamlit

- Built a scalable **LLM-powered analytics platform** using **RAG pipelines**, embeddings, and semantic retrieval workflows to analyze enterprise review datasets, reducing insight-generation time from **4 hours to minutes** while enabling automated trend analysis.

Two-Stage Recommendation System (MovieLens 32M) | Python, PyTorch, LightGCN, DeepFM, MLflow

- Architected a scalable **two-stage recommendation pipeline** combining LightGCN retrieval and DeepFM reranking with leakage-aware temporal validation, optimizing **NDCG@10** and **Precision@10** through iterative experimentation and MLflow evaluation workflows.

A/B Testing and Uplift Modeling System (Criteo Ad Experiment) | Python, causalml, scikit-uplift, statsmodels

- Built an experimentation-focused **uplift modeling pipeline** using causal inference, hypothesis testing & randomized advertising datasets to identify high-response customer segments, demonstrating a **65%** relative conversion lift supporting targeted promotional strategies.
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Education

Southern Methodist University, Dallas, TX

M.S. in Computer Science—Artificial Intelligence and Machine Learning

May 2025

GPA: 3.5/4